

電腦會聽你的指令嗎？

Will the computer follow your instructions?

```
message = "Hello, AI!"  
print(message)
```

猜猜看：螢幕會顯示甚麼？

Guess: what will appear on the screen?

先看老師示範，開啟自己的副本後再試

Watch the teacher first, then repeat in your own copy

換一句話，試試看

Change the greeting and try again

```
message = "Hello, future robot!"  
print(message)
```

改引號內的文字，再執行一次

Change the text inside the quotes, then run again

成功了？你已經完成第一個 Python 小實驗

Did it work? You have completed your first Python experiment

今天的挑戰：讓電腦找一條線

Today's challenge: let the computer find the line

上堂：你用眼睛觀察，畫一條合適的直線

Last time: you looked at the dots and drew a line

今堂：用 Python 執行、改數據、作預測

Today: use Python to run code, change data and predict

目標是看懂變化，無需背下每一行程式

Notice what changes. You do not need to memorise every line

Python、Notebook、Colab 是甚麼？

What are Python, a notebook and Colab?

Python：給電腦指令的程式語言

Python: the programming language for instructions

Jupyter Notebook：把文字、程式和結果放在一起

Jupyter Notebook: text, code and results together

Colab：在瀏覽器使用 Notebook，程式在雲端執行

Colab: use a notebook in a browser; code runs on a cloud computer

你的第一本「實驗簿」

Your first experiment notebook

開啟老師的連結，儲存自己的副本
Open the teacher's link and save your own copy

文字格是說明；程式格是可執行的指令
Text cells explain; code cells contain instructions to run

由上至下執行。結果會出現在格子下方
Run from top to bottom. Results appear below each cell

資源 / Resources: gordany.github.io/f2-linear-regression/workshop-2/

數據也要排好隊

The data needs matching pairs

```
hours    = [0, 1, 2, 3, 4, 5, 6]  
scores  = [53, 46, 59, 52, 63, 54, 65]
```

第 2 對：(1 小時, 46 分)。兩個列表要有相同數量

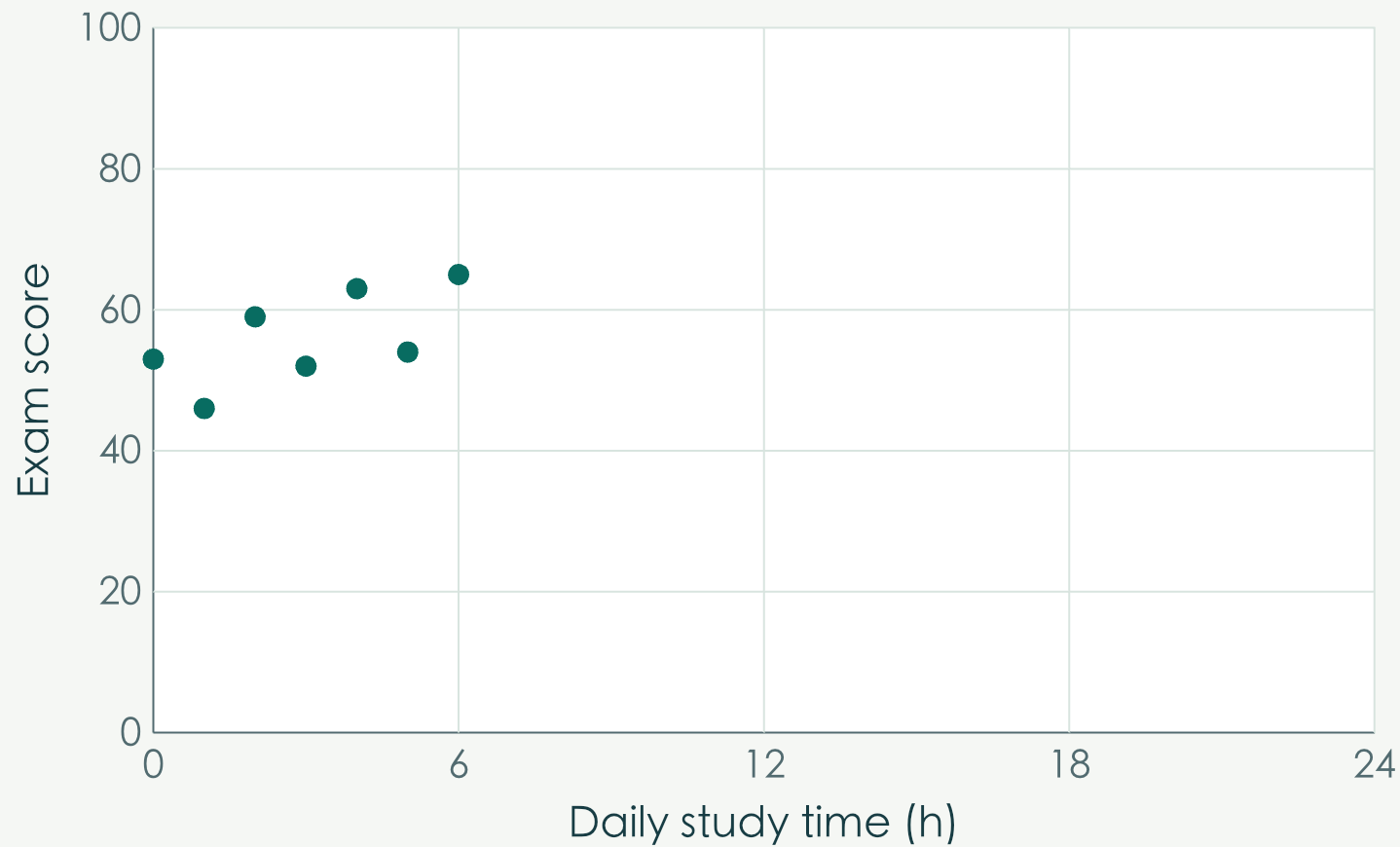
The second pair is (1 hour, 46 marks). Both lists need equal lengths

先看散點，猜每天 3.5 小時會得多少分

Look at the dots first. Guess the score for 3.5 hours a day

先猜一猜，暫時不揭曉

Make a guess before the reveal



3.5 小時，大約幾分？

About what score at 3.5 hours?

先寫答案，再看模型

Write your guess before the model

fit：讓工具從例子找直線

fit: let the tool learn a line from examples

```
model = LinearRegression()  
model.fit(X, y)
```

X 是整理好的輸入，y 是相應成績

X contains prepared inputs; y contains their scores

直線概括整體趨勢，不必穿過每一點

The line summarises the trend; it need not pass through every dot

predict : 請模型猜一個新情況
predict: ask the model about a new case

```
study_hours = 3.5  
model.predict([[study_hours]])
```

只改 study_hours : 試 3.5 , 再試 10

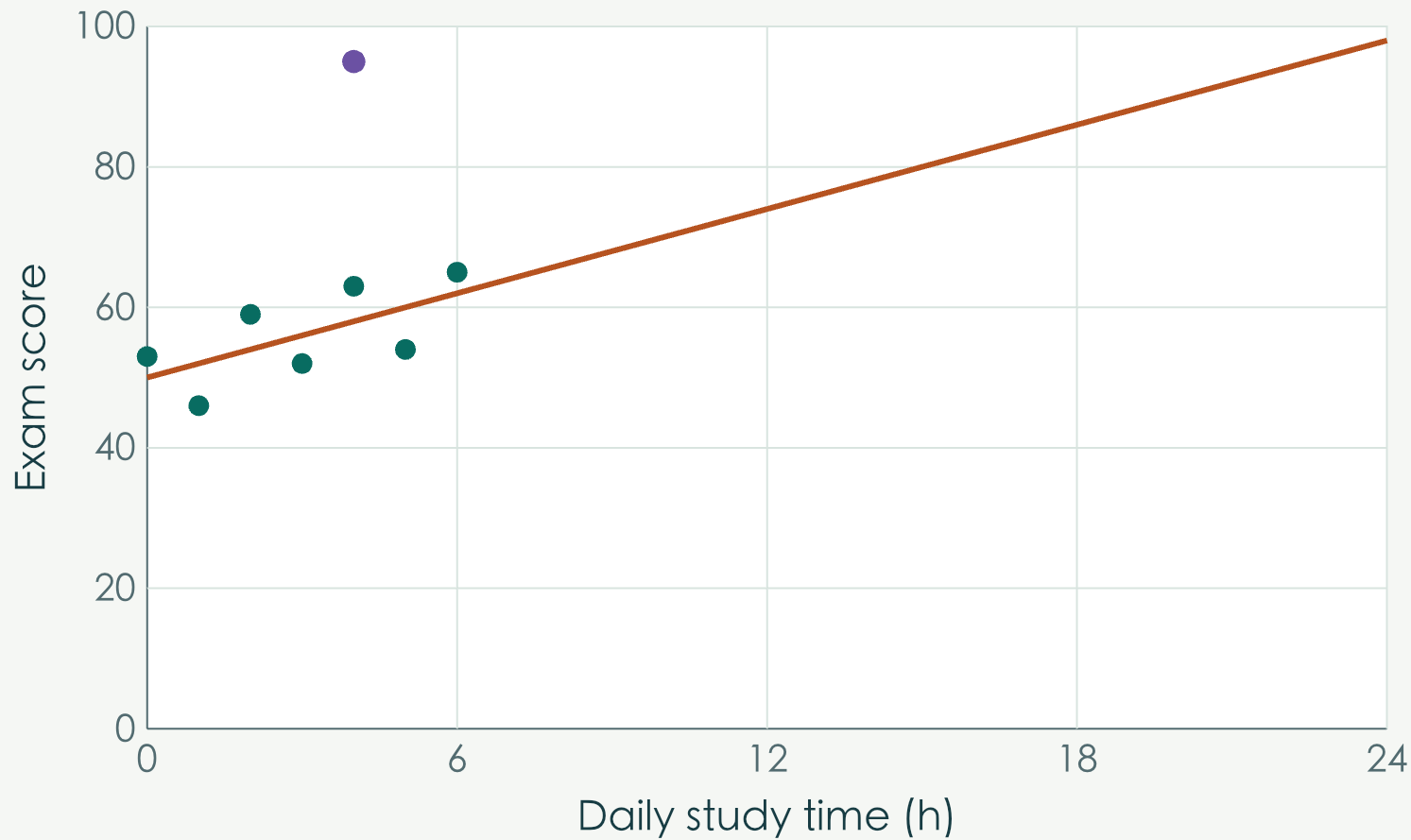
Change only study_hours: try 3.5, then 10

預測是估計。10 小時超出原來 0–6 小時的觀察範圍

A prediction is an estimate. 10 hours is outside the observed 0–6 range

一個奇怪的新點，會搗亂嗎？

Can one surprising point change the line?



```
new_point=[4,95]
```

先猜，再執行

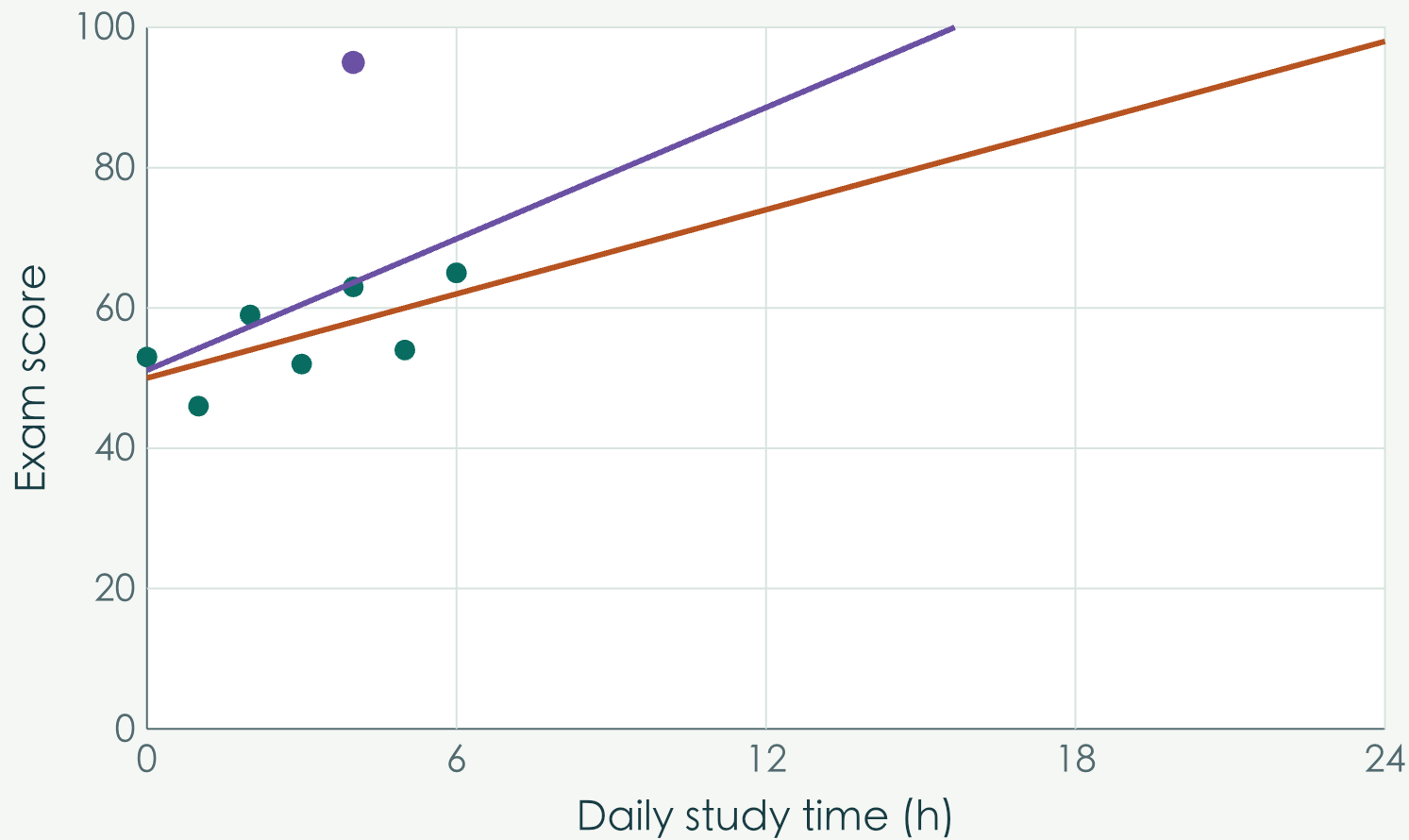
Predict, then run

直線會怎樣變？

How might the line change?

揭曉：原來和加點後的直線

Reveal: compare the two lines



`new_point=[4,95]`

3.5 小時的預測 / At 3.5 hours

原來約 57 分

Before: about 57 marks

加點後約 62 分

After: about 62 marks

橙：原來 / Orange: before 紫：加點後 / Purple: after

錯誤訊息是提示，不是評分

An error message is a clue, not a grade

先看最後一行提示，再檢查是否漏了逗號或引號

Read the last clue. Check commas and quotes

改過原數據？先重跑數據和建立模型的格子

Edited the original data? Rerun the data and model cells first

不必亂改。先復原剛才的改動，或請同伴／老師幫忙

Avoid random edits. Undo your last change or ask for help

這和 AI 有甚麼關係？

How does this connect to AI?

我們提供例子，工具從例子建立模型

We provide examples; the tool fits a model from them

改變數據，模型的預測可能改變

Change the data, and predictions may change

程式能執行，不代表模型的答案一定正確

Working code does not guarantee a correct prediction

把你的發現交給下一位偵探

Share your discovery with another detective

你成功改了甚麼？結果有甚麼變化？

What did you change successfully? What happened?

你較信賴哪個預測？為甚麼？

Which prediction do you trust more? Why?

下一堂：讓電腦猜「是哪一類」，而不是「幾多分」

Next time: predict which class, instead of how many marks